NB34: CFTR Feature Engineering

Tarih: 2026-06-23 23:27

1. Motivasyon

NB20 S0c_COMBINED modeli FE olmadan LOO-MCC=0.644, Boot-F1=0.863, Precision=1.0 elde ediyor.
Literatur arastirmasi (FCS, CFTR-MetaPred, AlphaMissense, PHACTboost) sonucu
16 feature (5 grup) belirlendi. Missingness (Grup C) CFTR'de anlamsiz (r<0.05).
Hedef: LOO-MCC = 0.644 -> 0.67-0.70 arasi.

2. Feature Gruplari

Grup A (3): FCS-tarzi frekans x konservasyon (fcs_ek9 r=-0.492)
Grup B (5): EK skor birlesimleri (ek_mean_top3 r=0.476)
Grup D (4): AA fizikokimyasal delta (hydro_abs r=0.170)
Grup E (4): AA substitusyon (grantham, blosum62, grantham_cat, charge_change)
Grup C: CIKARILDI -- tek-gen panelinde missingness homojen

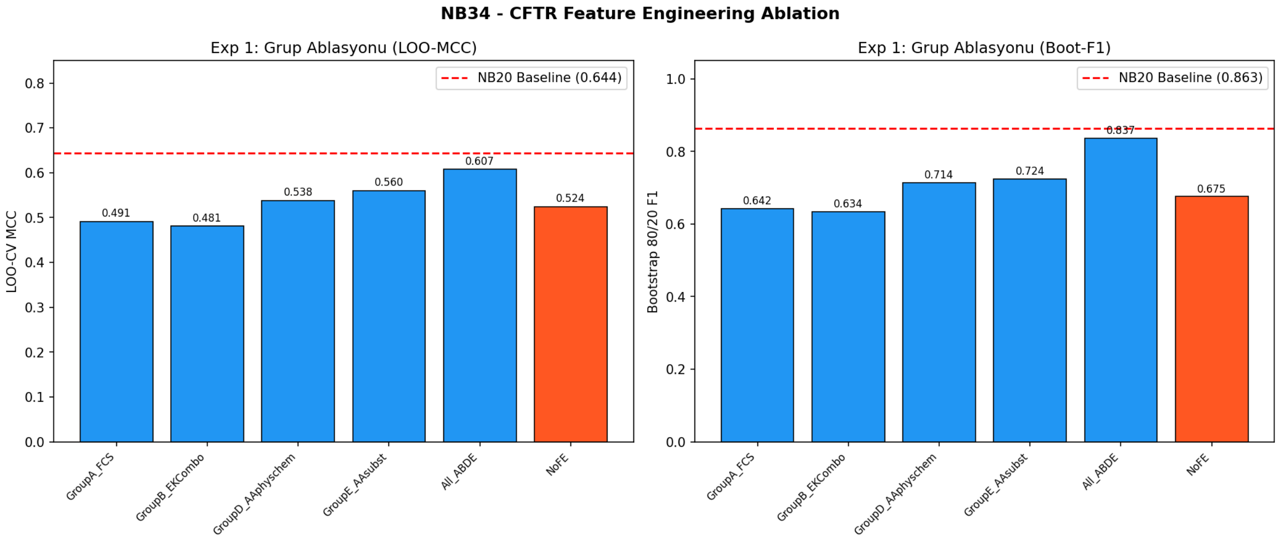
3. Sonuclar

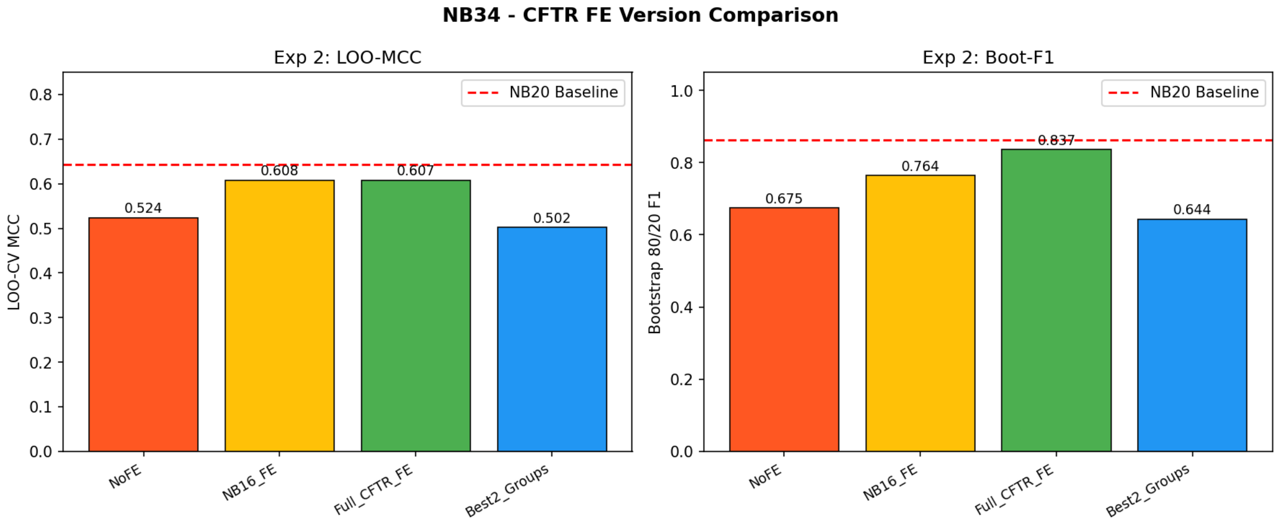
Deney	LOO-MCC	Boot-F1	Std	Prec	FP	FN
E1a_GroupA_FCS	0.4913	0.6423	0.1106	0.970	2	26
E1b_GroupB_EKCombo	0.4808	0.6343	0.1286	0.969	2	27
E1c_GroupD_AAphyschem	0.5381	0.7139	0.1256	0.985	1	25
E1d_GroupE_AAsubst	0.5602	0.7240	0.1339	0.985	1	23
E1e_All_ABDE	0.6074	0.8367	0.1136	1.000	0	22
E1f_NoFE	0.5243	0.6754	0.1269	0.971	2	23
E2a_NoFE	0.5243	0.6754	0.1269	0.971	2	23
E2b_NB16_FE	0.6081	0.7644	0.1198	0.986	1	19
E2c_Full_CFTR_FE	0.6074	0.8367	0.1136	1.000	0	22
E2d_Best2_Groups	0.5020	0.6438	0.1169	0.970	2	25

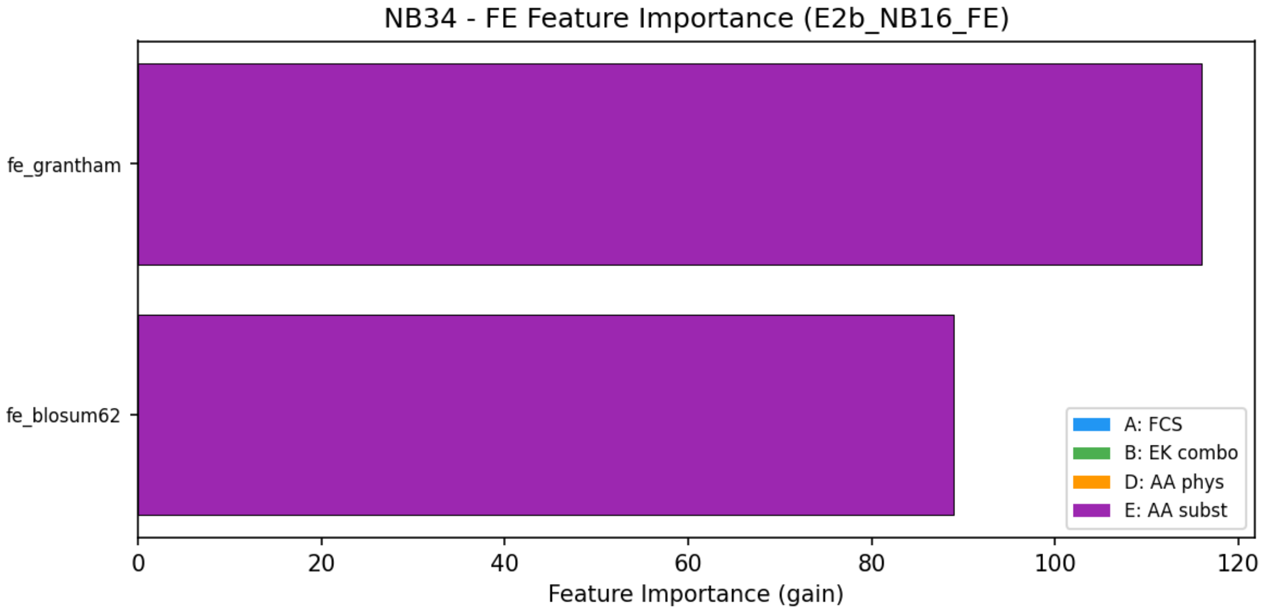
En iyi model: E2b_NB16_FE
LOO-MCC: 0.6081 (delta=-0.0355 vs NB20)
Boot-F1: 0.7644 +/- 0.1198
CM: TN=20, FP=1, FN=19, TP=71

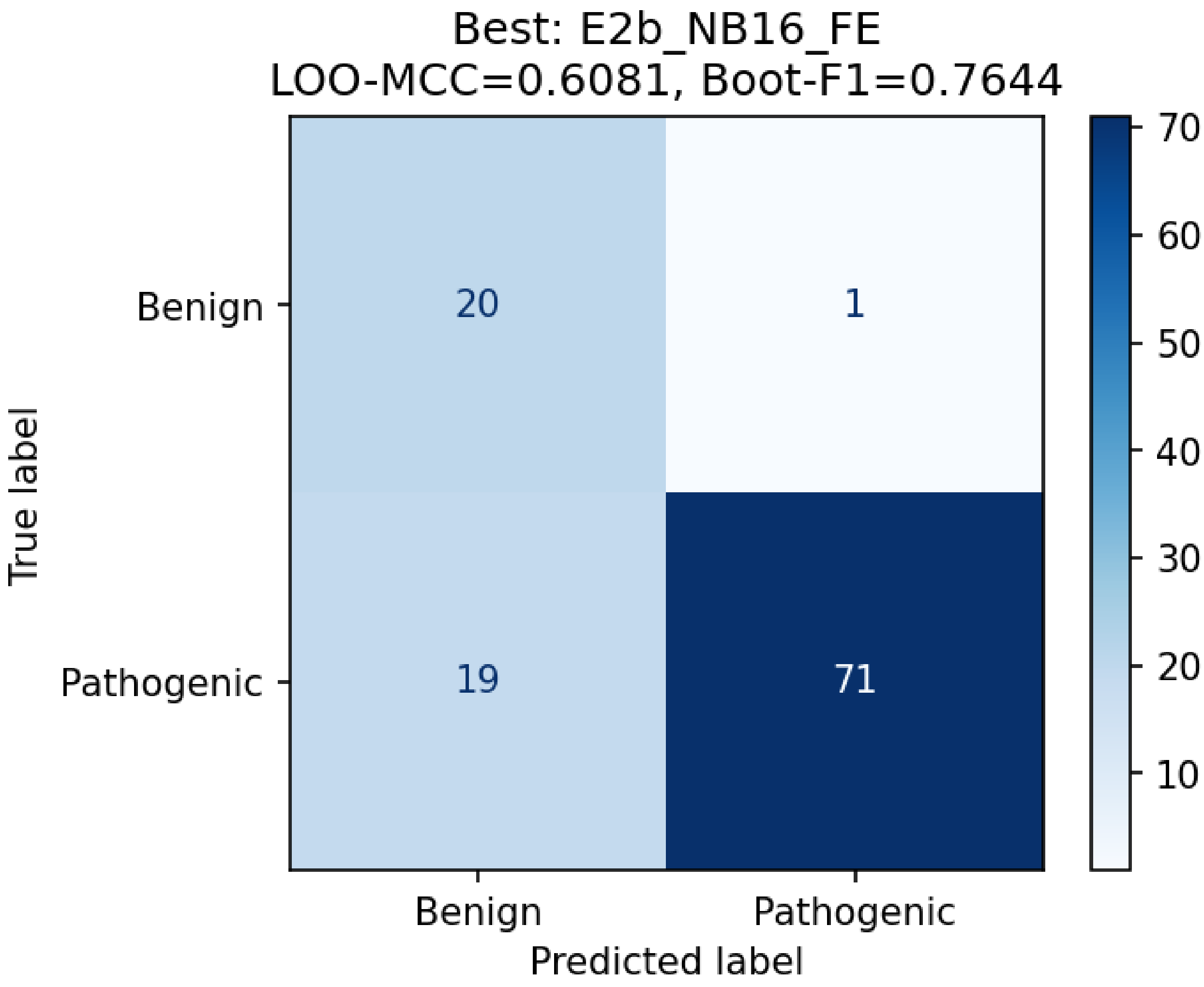
4. Gorseller

NB34 - CFTR Feature Engineering Ablation









5. Yorum ve Karar

FE CFTR'de deger katmadi (-0.0355). NB20 baseline korunuyor.

CFTR'de n=21 benign nedeniyle MCC farki > 0.05 olmadan 'kazanan' ilan edilmez.
Boot-CI=[0.00-1.00] nedeniyle Boot-F1 TEK BASINA karar kriteri DEGIL.